

PRODUCT OVERVIEW

MethanolX360 is an autonomous, closed-loop predictive decision-support and digitisation platform engineered specifically for the integrated Methanol Complex. Streaming 24/7 DCS and plant historian telemetry is continuously ingested, reconciled for mass/energy balance, and evaluated by hybrid first-principles chemical thermodynamics paired with neural surrogate AI digital twins. The system continuously benchmarks the Steam Methane Reformer (SMR), Synthesis Reactor Loop, and Distillation Train against dynamic optimal operational envelopes, detects hidden kinetic and hydraulic constraints, and serves ranked, safety-constrained operator setpoints directly to board consoles and APC supervisors to maximize methanol yield, reduce specific energy consumption, and prolong catalyst life.

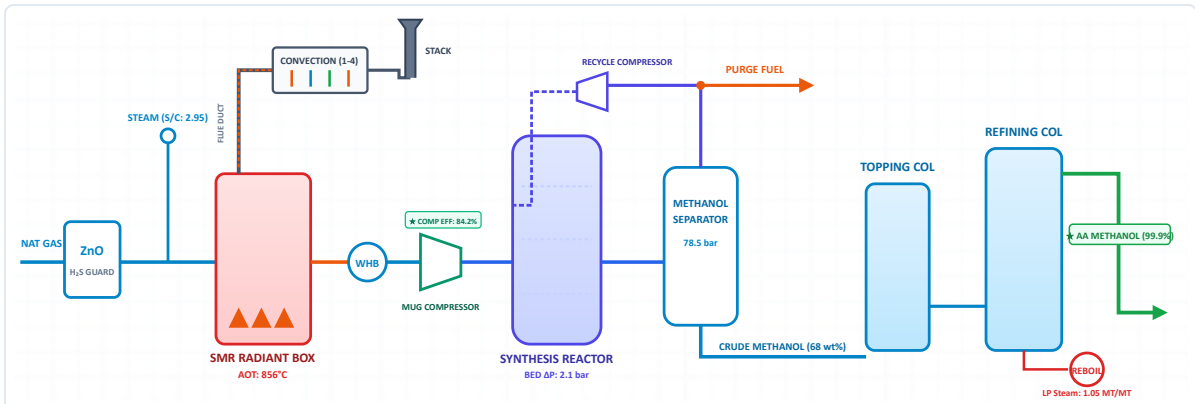
01 Unit Configuration & Ingestion

Shown: equipment configuration, then model selection against the assets.

Asset Hierarchy	Primary Reformer (SMR), Syngas Compression, Synthesis Loop (Quench/BWR), Distillation Train.
Reformer Geometry	Top-fired radiant box (For Ex: 480 catalyst tubes, 4 chambers), dual convection pass, steam drum & WHB.
Synthesis Loop	Multi-bed radial quench converter & BWR, Methanol Separator, recycle turbine compressor.
Distillation Train	3-column train: Topping column (light ends), HP/LP refining columns, reboiler thermosiphons.
Operating Modes	Full-rate, NG-constrained, high-CO ₂ import stoichiometry, catalyst aging (EOR) mode.
Design Envelope	Arch Exit Temp - (For Ex: 856°C), WHB high-pressure steam generation, S/C ratio (For Ex: 2.95 ± 0.05).
Synthesis Kinetics	Nameplate capacity (For Ex: 1,850 TPD AA-grade), bed ΔP Ergun framing, peak hotspot (For Ex: < 265°C).
Product Quality	AA-Grade Methanol (For Ex: > 99.85 wt%), Light Ends (For Ex: < 0.12%), Ethanol slip (For Ex: < 10 ppm).

02 Methanol Complex Overview

End-to-end 2D process flow: SMR, Syngas, Loop & Distillation



03 AI-Powered Model Agent Suite

12 specialized domain & neural agents running 24/7 on live plant data

Proven in Production: Digital Models Integrated into our Methanol Complex Solution

Fusing rigorous first-principles chemical engineering thermodynamics with physics-informed neural network (PINN) surrogates, the platform continuously tracks asset health, computes unmeasured KPIs, and drives autonomous closed-loop advisory.

Primary Reforming & Convection

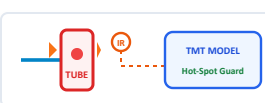
3 AI MODELS

Clean TMT & Tube Health Monitor

Arch Limit: 856°C Hot-Spot Guard: Active Tube Life Protected

Detects localized furnace radiant tube hot spots and balances burner firing to prevent premature tube rupture.

Physics-Informed NN



Reformer Methane Conversion Optimizer

Methane Conv: 84.8% S/C Target: 2.95 Carbon Guard: Safe

Maximizes methane conversion while dynamically managing steam injection to safeguard catalyst from carbon fouling.

Kinetics Equilibrium



Desulfurizer Bed Saturation & Life

Sulfur Slip: <0.02 ppm Bed Life: 180 Days Guard Active

Tracks zinc oxide sulfur absorption kinetics and forecasts remaining bed days to shield reforming catalyst.

Adsorption Kinetics



Distillation Train & Purity Assurance

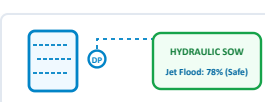
3 AI MODELS

Distillation Hydraulic Safe Window (SOW)

Jet Flooding: 78% (Safe) Weep Guard: Active Turndown Safe

Real-time tray hydrodynamic rating establishing safe operating bounds to prevent weeping and jet flooding.

Hydrodynamic Rating



Column Feed Rate & Reflux Optimizer

Purity: 99.85% AA Reboiler Steam: Min Reflux Tuned

Determines optimal column feed rates and reflux ratios to guarantee IMPCA AA export purity at lowest steam boilup.

Nonlinear Optimization



Exchanger Fouling Index

Cleanliness: 91% Heat Recovery: High Wash Alert: None

Monitors heat transfer coefficients across feed preheaters and saturator packing to schedule proactive washes.

Dynamic HTC Tracking



Syngas & Methanol Synthesis Loop

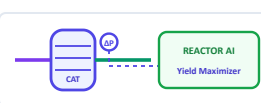
3 AI MODELS

Synthesis Reactor Bed Health & ΔP

Methanol Yield: Max Bed ΔP: Normal Peak Temp: Safe

Tracks catalyst bed reaction front migration and pressure drop to sustain maximal single-pass methanol yield.

Heterogeneous Kinetics



Syngas Compressor & Turbine Health

Polytropic Eff: 82.4% Surge Margin: +14% Steam Saved

Optimizes polytropic efficiency, surge margins, and high-pressure steam governor control across feed gas machines.

Polytropic Efficiency



Methanol Separator Recovery Maximizer

Purge Loss: <0.3% Loop Pressure: Stable Yield Recovered

Rigorous high-pressure vapor-liquid equilibrium flash modeling minimizes valuable methanol carryover in fuel purge gas.

Rigorous VLE Flash



Site Energy, Emissions & GenAI

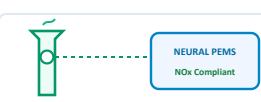
3 AI MODELS

Stack Emissions

NOx Compliance: 100% Excess O₂: Balanced Virtual Analyzer

Neural virtual analyzer forecasting flue gas NOx emissions and excess O₂ for continuous environmental compliance.

Virtual Sensor



Complex Energy & Steam Network

Specific Energy: -3.2% Steam Headers Balanced Fuel Saved

Optimizes complex-wide energy consumption by balancing high, medium, and low pressure steam headers against power.

Mixed-Integer LP



Autonomous GenAI Operations Copilot

Automated Shift Handover Root-Cause Diagnostics Digital SOPs

Delivers plain-language shift summaries, root-cause anomaly diagnostics, and standard operating procedures to operators.

Domain LLM



04 Opportunity and Operator Actions

Real-time benchmarking · Multi-variable historian trend gaps · Safety-constrained advisory

OPPORTUNITY OVERVIEW

LIVE ADVISORY

AI PRODUCTION OPTIMIZER

METHANOL PRODUCTION OPPORTUNITY

Selected Cluster: Cluster 4 (Peak Capacity Matched) · Refreshed: 16:47:42

CURRENT PRODUCTION

1,755.58 TPD

Operating Baseline

BENCHMARK PRODUCTION

1,841.46 TPD

Demonstrated High-Yield State

POTENTIAL GAIN

+85.88 TPD

Realizable Delta

IMPROVEMENT

+4.89%

Synthesis Uptick

RELIABILITY

98% (High)

Data Confidence

AUTONOMOUS PROCESS OPTIMIZATION RECOMMENDATIONS (2 SELECTED)

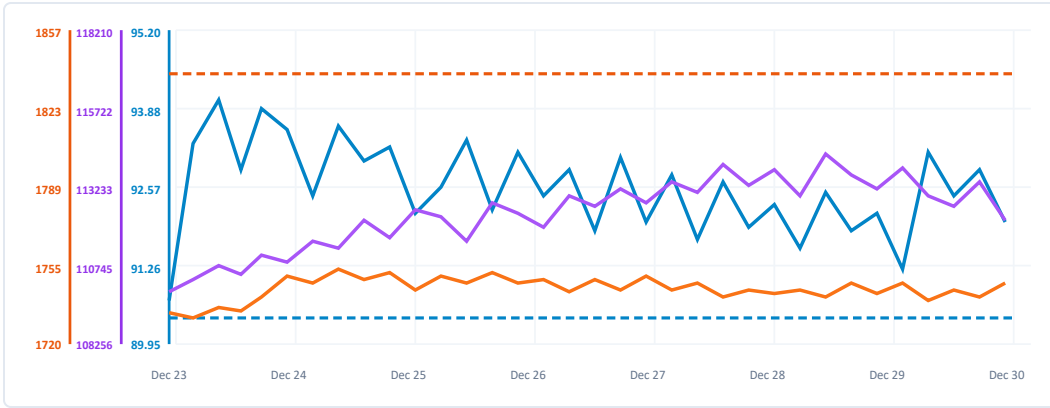
Closed-loop actionable recommendations with process safety constraints

PRIORITY	PARAMETER	ACTUAL	OPTIMUM	IMPACT	RECOMMENDED ACTION
MEDIUM	LOW LOOP PRESSURE Loop Pressure Control	85.75	94.36	+4.58 TPD	Actual loop pressure (85.75 kg/cm ² g) < optimum (94.36). Suggest increasing in steps of 1.0 kg/cm²g towards optimum while monitoring compressor head.
LOW	LOW SYNGAS FLOW Fresh Syngas Compressor Feed	255.50	265.82	+42.69 TPD	Actual syngas feed flow (255.50 kNm ³ /h) < optimum (265.82). Action temporarily deferred: SMR reformer load was recently increased by +4.5% (currently operating at 98.3% of design limit). Dynamic protocol requires monitoring tube skin temperatures and draft for 4 days of thermal stabilization; if all furnace parameters remain safely within limits, proceed with further incremental flow ramping to capture +42.69 TPD.

PROCESS VARIABLE TREND ANALYSIS

Multi-variable historian telemetry with dynamic optimum targets

1D 1W 2W 1M



Carbon Yield MUG Flow Methanol Prod Target

CONTRIBUTORS — PROCESS (4 SELECTED)

Top process opportunity impacts

PARAMETER	ACTUAL	OPTIMUM	CONTRIBUTION	IMPACT	STATE
FRESH SYNGAS FEED FLOW	255.50	265.82	68.05%	-42.69 TPD	●
M VALUE	3.01	2.97	20.67%	+12.97 TPD	●
LOOP PRESSURE (KG/CM ² G)	85.75	94.36	7.28%	-4.58 TPD	●
BED 2 DELTA T	36.48	29.10	2.48%	-1.56 TPD	●

Basis: Live benchmarking — Methanol Synthesis Loop (Cluster 4: Peak Capacity). Production: 1,755.58 TPD vs Benchmark: 1,841.46 TPD (+85.88 TPD / +4.89%). Loop pressure incremented in 1.0 kg/cm²g steps; MUG constrained by upstream SMR load at 98.3%.

DELIVERY

Configured, validated, and commissioned by Ingenero chemical and process automation engineers.

FLEXIBILITY

Natural gas or rich gas feeds; configurable catalyst kinetics; customizable KPI & virtual analyzer suite.

SCALE

Multi-train complexes reuse configuration with zero or minimal rework per operating train.

Contact for a trial:
marketing@ingenero.com